## Appendix A

# MATHEMATICAL TOOLS FOR PCRA

**Planned for publication in 2026 by R. Douglas Martin, Thomas K. Philips, Bernd Scherer and Kirk Li.** **All rights reserved. © Copyright 2025**

10/12/25

# Contents


**A MATHEMATICAL TOOLS FOR PCRA** . . . . . 1

A.1 Vectors and Vector Spaces . . . . . 5

A.1.1 Vectors . . . . . 5

A.1.2 The Distance Between Two Vectors . . . . . 6

A.1.3 Vector Spaces . . . . . 7

A.1.4 Projections of a Vector Space . . . . . 9

A.2 Matrices and Matrix Arithmetic . . . . . 10

A.2.1 Some Basic Definitions . . . . . 10

A.2.2 The Rank of a Matrix . . . . . 10

A.2.3 Matrix Addition and Multiplication . . . . . 11

A.2.4 Some Special Matrices . . . . . 13

A.2.5 Toeplitz Matrices . . . . . 14

A.2.6 Block Matrices . . . . . 14

A.2.7 Echelon Form and LU Factorization . . . . . 16

A.2.8 The Determinant of a Matrix . . . . . 16

A.2.9 The Inverse of a Matrix . . . . . 19

A.2.10 Inverses of Special Matrices . . . . . 20

A.2.11 The Sherman–Morrison–Woodbury Matrix Inversion Lemma . . . . . 22

A.2.12 The Inverse of a Covariance Matrix . . . . . 23

A.2.13 The Mahanalobis Distance . . . . . 25

A.2.14 Eigenvalues and Eigenvectors . . . . . 25

A.2.15 Positive Definite Matrices . . . . . 28

A.2.16 QR Decomposition and the Gram–Schmidt Procedure . . . . . 29

A.2.17 The Singular Value Decomposition . . . . . 31

A.3 Some Results on Functions . . . . . 33

A.3.1 Derivatives of Functions . . . . . 33

A.3.2 Taylor Series . . . . . 34

A.3.3 Convex and Concave Functions . . . . . 35

A.3.4 Multivariate Convex and Concave Functions . . . . . 36

A.3.5 Euler's Theorem for Homogeneous Functions . . . . . 36

A.4 Optimization . . . . . 37
A.4.1 The General Problem of Optimization . . . . . 37
A.4.2 Optimizing Twice–Differentiable Convex Functions . . . . . 37
A.4.3 The Method of Lagrange Multipliers . . . . . 38

This Appendix contains a variety of basic mathematical results that prove useful in the context of portfolio construction and risk management. We expect that much of the material provided here will be used primarily to refresh the reader's memory, thereby facilitating its use throughout this book. Our treatment of matrices and Linear Algebra owes a great deal to Gentle (2017) and Strang (2016), and we strongly recommend both books to anyone with an interest in this under rated field. In addition, we refer readers who have an interest in optimization to Luenberger and Ye (2008).

## A.1 Vectors and Vector Spaces

### *A.1.1 Vectors*

A *vector* is a collection of objects that supports two arithmetic operations: vector addition (denoted $+$) and scalar multiplication (denoted $\cdot$). It is useful to think of a vector as a list of $n$ real numbers that correspond to a point in $n$–dimensional Euclidean space with $n$ orthogonal axes, as this interpretation clearly supports elementwise addition provided both vectors have the same number of elements, i.e., if

$$\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \tag{A.1}$$

then

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \tag{A.2}$$

$$= \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}. \tag{A.3}$$

In addition, it supports scalar multiplication according to

$$c \cdot \mathbf{u} = \begin{bmatrix} c\,u_1 \\ \vdots \\ c\,u_n \end{bmatrix}. \tag{A.4}$$

However, not all vectors are defined in this way. In particular, polynomials and square–integrable functions of one real variable are also vectors, as both support vector addition[1] and scalar multiplication.

One type of vector that we repeatedly encounter in this book is a vector of *security returns*, each element of which is lower bounded by $-100\%$. Another is a vector of *security weights* associated with a portfolio. These weights may be *constrained* (e.g. the long–only constraint requires all weights to be positive and

[1] The sum of two polynomials is also a polynomial, and the same is true for square–integrable functions of one real variable.

to sum to 1) or *unconstrained* (i.e. the weights may be positive or negative, and may not sum to 1, as is often the case when a portfolio is levered).

Vectors also support vector multiplication via the *dot product,* or *inner product*[2]. The dot product of two vectors of equal length, $\mathbf{u}$ and $\mathbf{v}$, is defined to be[3]

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i. \tag{A.5}$$

More generally, the inner product of $\mathbf{u}$ and $\mathbf{v}$ is written as $\langle \mathbf{u}, \mathbf{v} \rangle$, and is a real number that satisfies the following criteria:

1. Non–negativity: $\langle \mathbf{u}, \mathbf{0} \rangle = 0 \; \forall \, \mathbf{u}$, and $\langle \mathbf{u}, \mathbf{u} \rangle > 0$ if $\mathbf{u} \neq \mathbf{0}$
2. Commutativity: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
3. Scalar multiplication: $\langle c\,\mathbf{u}, \mathbf{v} \rangle = c \, \langle \mathbf{u}, \mathbf{v} \rangle$ for all scalars $c$
4. Distributivity: $\langle \mathbf{u} + \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle$

Two vectors, $\mathbf{u}$ and $\mathbf{v}$, are said to be *orthogonal* if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0. \tag{A.6}$$

The *norm* of a vector is the square root of its inner product with itself.

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}. \tag{A.7}$$

In Euclidean space, the norm of a vector is equal to its length.

### *A.1.2 The Distance Between Two Vectors*

The vector

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \tag{A.8}$$

corresponds to the point $x = 1$, $y = -2$ in the $X - Y$ plane. We can apply Pythagoras' Theorem to compute its norm, or the distance from the origin to the point $(1, -2)$:

$$\begin{aligned} \|\mathbf{u}\| &= \sqrt{1^2 + (-2)^2} \\ &= \sqrt{5}. \end{aligned} \tag{A.9}$$

[2] The term dot product is typically used when the vector space under consideration is $n$–dimensional Euclidean space. In more general vector spaces, the term inner product is more commonly used. There is another type of vector product, the *cross product*, which sees application mainly in physics. We ignore it here.

[3] This definition of an inner product can be extended to cover complex numbers as well, but we restrict our attention to real numbers as complex numbers are exceedingly rare in portfolio construction.

In general, the norm of a vector with $n$ elements in Euclidean space is

$$\|\mathbf{u}\| = \sqrt{u_1^2 + \cdots + u_n^2}, \tag{A.10}$$

and this concept is easily generalized to obtain the *distance* between two vectors:

$$\begin{aligned} d_{\mathbf{u},\mathbf{v}} &\triangleq \|\mathbf{u} - \mathbf{v}\| \\ &= \sqrt{(u_1 - v_1)^2 + \cdots + (u_n - v_n)^2}. \end{aligned} \tag{A.11}$$

It is important to note that this definition of distance is not unique: the distance between two square–integrable functions of one real variable, $f(x)$ and $g(x)$, for example, can be written as

$$d_{f,g} = \sqrt{\int (f(x) - g(x))^2 \, dx}. \tag{A.12}$$

Regardless of how it is defined, a valid measure of distance must satisfy three properties:

1. Non-negativity: For all vectors $\mathbf{u}, \mathbf{v}$, $d_{\mathbf{u},\mathbf{v}} \geqslant 0$.
2. Symmetry: For all vectors $\mathbf{u}, \mathbf{v}$, $d_{\mathbf{u},\mathbf{v}} = d_{\mathbf{v},\mathbf{u}}$.
3. Triangle Inequality: For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$, $d_{\mathbf{u},\mathbf{w}} \leqslant d_{\mathbf{u},\mathbf{v}} + d_{\mathbf{v},\mathbf{w}}$.

The triangle inequality get its name from its geometric interpretation: the sum of the lengths of any two sides of a triangle must be at least as large as its third side. When the vectors in question are lists of real numbers, and the measure of distance between vectors is the Euclidean distance, the triangle inequality is equivalent to the Cauchy–Schwarz inequality for real numbers:

$$\left( \sum_{i=1}^{n} u_i v_i \right)^2 \leqslant \sum_{i=1}^{n} u_i^2 \cdot \sum_{i=1}^{n} v_i^2, \tag{A.13}$$

with equality if and only if $v_i = c\, u_i$ for some constant $c$.

While proofs of these inequalities abound, we recommend reviewing their treatment in Steele (2004), which is both comprehensive and accessible to the non–specialist. The dot product admits an equivalent, geometric, definition:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta, \tag{A.14}$$

where $\theta$ is the angle between $\mathbf{u}$ and $\mathbf{v}$. Orthogonality is equivalent to condition that $\theta = 90^\circ$.

### *A.1.3 Vector Spaces*

A *vector space* adds structure to sets of vectors and the arithmetic operations associated with them. A vector space $\mathcal{S}$ has the following properties:

1. Additive Closure: For all vectors $\mathbf{u}, \mathbf{v} \in \mathcal{S},\ \mathbf{u} + \mathbf{v} \in \mathcal{S}$.
2. Multiplicative Closure: For all vectors $\mathbf{u} \in \mathcal{S}$, and for all scalars $c,\ c \cdot \mathbf{u} \in \mathcal{S}$.
3. Additive commutativity: For all vectors $\mathbf{u}, \mathbf{v} \in \mathcal{S},\ \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ .
4. Distributivity with respect to scalar and vector addition: For all vectors $\mathbf{u}, \mathbf{v} \in \mathcal{S}$, and for all scalars $c, d,\ (c + d) \cdot (\mathbf{u} + \mathbf{v}) = c \cdot \mathbf{u} + c \cdot \mathbf{v} + d \cdot \mathbf{u} + d \cdot \mathbf{v} \in \mathcal{S}$.
5. Additive Associativity: For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{S},\ (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
6. Multiplicative Associativity: For all vectors $\mathbf{u} \in \mathcal{S}$, and for all scalars $c, d,\ \ (c\,d) \cdot \mathbf{w} = c \cdot (d \cdot \mathbf{w})$.
7. Existence of additive and multiplicative identities. There exists a vector $\mathbf{0} \in \mathcal{S}$ and a scalar 1, such that for all $\mathbf{u} \in \mathcal{S},\ \mathbf{u} + \mathbf{0} = \mathbf{u}$ and $1 \cdot \mathbf{u} = \mathbf{u}$.
8. Existence of an additive inverse. For all vectors, $\mathbf{u} \in \mathcal{S}$, there exists a vector $-\mathbf{u} \in \mathcal{S}$ such that $\mathbf{u} + -\mathbf{u} = \mathbf{0}$.

It is important to check that each of these properties is satisfied before declaring a set of vectors to be a vector space. For example, if $\mathbf{u}$ is a vector of security returns, and if $c$ is a real number, and if vector addition and scalar multiplication are defined elementwise and follow the normal rules of arithmetic, $c \cdot \mathbf{u}$ does not always satisfy a basic property of returns: they cannot be smaller than $-100\%$. If, on the other hand, $\mathbf{u}$ is a vector of *logarithmic returns* (see Chapter 3 for a discussion of various types of returns) it is easy to see that $\mathcal{S}$ is a vector space.

An *inner product space* is a vector space in which the inner product is well defined.

A *subspace* of a vector space is a subset of $\mathcal{S}$ that satisfies all the requirements of a vector space. In particular, vector addition and scalar multiplication must always result in a vector that lies in the subspace.

For example, let $\mathcal{S}_{XY}$ be the set of all vectors with 2 real components, i.e. vectors in the $X$–$Y$ plane, with elementwise vector addition and scalar multiplication. $\mathcal{S}_Y$, the subset of vectors whose first component is 0 (i.e. vectors that lie on the $Y$ axis), is a subspace of $\mathcal{S}_{XY}$, as is $\mathcal{S}_X$, the subset of vectors whose second component is 0 (i.e. vectors that lie on the $X$ axis).

Formally, a set of vectors $\mathbf{v}_{1,}, \mathbf{v}_2, \cdots, \mathbf{v}_k$ which are not all equal to $\mathbf{0}$ is said to be linearly independent if

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0} \Longleftrightarrow c_1 = c_2 = \cdots = c_k = 0. \tag{A.15}$$

The *dimension* of a vector space is the maximum number of linearly independent vectors contained in it, and the dimension of a subspace cannot exceed that of its parent space.

A set of linearly independent vectors $\mathcal{B}$ is said to be a *basis* for $\mathcal{S}$, or to *span* $\mathcal{S}$, if any vector $\mathbf{v} \in \mathcal{S}$ can be written as a unique linear combination of the vectors in $\mathcal{B}$. Bases are not unique, though one basis may prove more convenient than another in a particular application. All bases must have the same number of elements, and this must equal the dimension of the vector space.

If the vectors in a basis are all orthogonal to each other, the basis is called an *orthogonal basis*. If, in addition, the length of each basis vector is 1, it is said to be an *orthonormal basis*.

It is easy to see why $\mathcal{S}_{XY}$ is a two dimensional vector space: there are no non–zero values of $c_1$ and $c_2$ such that

$$c_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{A.16}$$

and every vector in the $X-Y$ plane can be represented as a unique combination of $c_1$ and $c_2$. It follows that

$$B = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \tag{A.17}$$

is an orthonormal basis for $\mathcal{S}_{XY}$[4].

Likewise, it is easy to see why $\mathcal{S}_Y$ is one–dimensional: for any two vectors whose first component is 0,

$$c_1 \begin{bmatrix} 0 \\ a \end{bmatrix} - \frac{ac_1}{b} \begin{bmatrix} 0 \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{A.18}$$

provided that $a$ and $b$ are not 0. Furthermore, any point on the $Y$ axis can be represented as a multiple of either vector. Exchanging the first and the second components of both vectors makes clear that the same argument applies to $S_X$.

### *A.1.4 Projections of a Vector Space*

Exact, but computationally demanding, calculations using vectors in a high dimensional vector space $\mathcal{S}_1$ are often approximated using representations of the vectors in a lower dimensional subspace $\mathcal{S}_2$. If $\mathcal{B}_2$ is an orthogonal basis for $\mathcal{S}_2$, the best approximate representation of a vector $\mathbf{u}_1 \in \mathcal{S}_1$ in $\mathcal{S}_2$ is its *projection*, $\mathbf{p}_1$, which is given by

$$\mathbf{p}_1 = \sum_{\mathbf{b}_i \in \mathcal{B}_2} \frac{\langle \mathbf{u}_1, \mathbf{b}_i \rangle}{\|\mathbf{b}_i\|^2} \mathbf{b}_i \tag{A.19}$$

In essence, the projection first captures, and then aggregates, all that $\mathbf{u}_1$ has in common with each of the orthogonal basis vectors of $\mathcal{S}_2$. The residual

$$\begin{aligned} \mathbf{r}_1 &= \mathbf{u}_1 - \mathbf{p}_1 \\ &= \mathbf{u}_1 - \sum_{\mathbf{b}_i \in \mathcal{B}_2} \frac{\langle \mathbf{u}_1, \mathbf{b}_i \rangle}{\|\mathbf{b}_i\|^2} \mathbf{b}_i \end{aligned} \tag{A.20}$$

must therefore be orthogonal to each basis vector, and must lie in $\mathcal{S}_2^{\perp}$, the *orthogonal complement* to $\mathcal{S}_2$[5]. Seen in this light, projecting a vector from $\mathcal{S}_{XY}$ to $\mathcal{S}_Y$ is equivalent to zeroing out its first component and preserving its second; the residual of this projection lies entirely in $\mathcal{S}_X$, which is the orthogonal complement to $\mathcal{S}_Y$.

[4] Likewise, the monomials $\{1, x, x^{2,}, \cdots\}$form a basis for the vector space of polynomials.

[5] The orthogonal complement of a subspace is a vector space comprised of all vectors that are orthogonal to the subspace.

## A.2 Matrices and Matrix Arithmetic

### A.2.1 Some Basic Definitions

We can stack $m$ vectors[6] with $n$ elements each side by side to form an $n \times m$ *matrix* as follows[7]:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_m \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{bmatrix}. \end{aligned} \tag{A.21}$$

If $n \neq m$, the matrix is said to be *rectangular*, and if $n = m$, it is said to be *square*. Rectangular matrices often arise when we model the returns of many securities using a few cross–sectional or time–series factors, while square matrices arise when representing the interaction between all possible pairs of securities or factors. A covariance matrix, for example, is square.

The transpose of a matrix $\mathbf{A}$, denoted $\mathbf{A}'$, is obtained by exchanging its rows and columns, i.e.

$$\mathbf{A}' = \begin{bmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & \vdots & \vdots \\ a_{1m} & \cdots & a_{nm} \end{bmatrix}. \tag{A.22}$$

Transposition transforms an $n \times m$ matrix into a $m \times n$ matrix.

### A.2.2 The Rank of a Matrix

The *column space* of a matrix is the vector space spanned by its columns, while its *row space* is the vector space spanned by its rows. The row space of a matrix is the column space of its transpose. The *null space* of a matrix, $\mathcal{N}(\mathbf{A})$, as well as its *left null space,* $\mathcal{N}(\mathbf{A}')$, are vector spaces that are orthogonal to its column and row spaces respectively. They are defined by the fact that vectors in the null space and left null space are mapped to $\mathbf{0}$ by $\mathbf{A}$ and $\mathbf{A}'$ respectively, i.e.

$$\mathcal{N}(\mathbf{A}) = \{\mathbf{x} \,|\, \mathbf{A}\mathbf{x} = \mathbf{0}\} \tag{A.23a}$$

$$\mathcal{N}(\mathbf{A}') = \{\mathbf{y} \,|\, \mathbf{A}'\mathbf{y} = \mathbf{0}\}, \tag{A.23b}$$

where the product of a matrix and a vector is defined in (A.29).

Closely related to the dimension of a vector space (the maximum number of linearly independent vectors contained in it) are the *row rank* of a matrix, or the maximum number of linearly independent

[6] We continue to require that all components of these vectors be real numbers, though much of this section can be generalized to complex numbers and finite fields.

[7] We follow the notation in Gentle (2017), and use $n$ to denote the number of rows and $m$ to denote the number of columns. Other authors exchange $n$ and $m$, but still put the number of rows before the number of columns.

*rows* contained in it, and its *column rank*, or the maximum number of linearly independent *columns* contained in it. Although one might think that the row and column ranks of a matrix will in general be different, it is surprisingly true that they are always equal. A proof of this non–intuitive result can be found in Gentle (2017). It follows that

$$\operatorname{rank}(\mathbf{A}_{n\times m}) \leqslant \min(n, m). \tag{A.24}$$

Furthermore

$$\operatorname{rank}(\mathbf{A}_{n\times m}) + \operatorname{rank}(\mathcal{N}(\mathbf{A}_{n\times m})) = m \tag{A.25a}$$
$$\operatorname{rank}(\mathbf{A}'_{m\times n}) + \operatorname{rank}\left(\mathcal{N}\left(\mathbf{A}'_{m\times n}\right)\right) = n, \tag{A.25b}$$

and

$$\operatorname{rank}(\mathbf{A} + \mathbf{B}) \leqslant \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}) \tag{A.26a}$$
$$\operatorname{rank}(\mathbf{AB}) \leqslant \min(\operatorname{rank}(\mathbf{A}), \operatorname{rank}(\mathbf{B})), \tag{A.26b}$$

where matrix addition and multiplication are defined in (A.27) and (A.30) respectively.

### *A.2.3 Matrix Addition and Multiplication*

We can add two matrices elementwise if they have the same numbers of rows and columns

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1m} + b_{1m} \\ \vdots & \vdots & \vdots \\ a_{n1} + b_{n1} & \cdots & a_{nm} + b_{nm} \end{bmatrix}, \tag{A.27}$$

and multiply a matrix by a scalar, just as we did with vectors:

$$c\cdot\mathbf{A} = \begin{bmatrix} c\,a_{11} & \cdots & c\,a_{1m} \\ \vdots & \vdots & \vdots \\ c\,a_{n1} & \cdots & c\,a_{nm} \end{bmatrix}. \tag{A.28}$$

A matrix may also be multiplied by a vector, to give rise to another vector. The product $\mathbf{Ax}$ is a *linear transformation* of the vector $\mathbf{x}$, and can also be viewed as a linear combination of the columns of the matrix, with the weights being given by the vector $\mathbf{x}$. The number of columns in $\mathbf{A}$ must equal the number of elements in $\mathbf{x}$ if the product is to be valid:

$$
\begin{aligned}
\mathbf{A}\mathbf{x} &= \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_m \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} \\
&= \begin{bmatrix} x_1\,\mathbf{a}_1 + \cdots + x_m\,\mathbf{a}_m \end{bmatrix} \\
&= \begin{bmatrix} a_{11}\,x_1 + \cdots + a_{1m}\,x_m \\ \vdots \\ a_{n1}\,x_1 + \cdots + a_{nm}\,x_m \end{bmatrix}.
\end{aligned} \tag{A.29}
$$

As a matrix is just a stacked set of vectors, (A.29) generalizes naturally to allow for the multiplication of two matrices that are conformable – the number of columns in the first matrix must equal the number of rows in the second:

$$
\begin{aligned}
\mathbf{A}\mathbf{B} &= \mathbf{A}\begin{bmatrix} \mathbf{b}_1 \cdots \mathbf{b}_p \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{A}\mathbf{b}_1 \cdots \mathbf{A}\mathbf{b}_p \end{bmatrix} \\
&= \begin{bmatrix} a_{11}\,b_{11} + \cdots + a_{1m}\,b_{m1} & \cdots & a_{11}\,b_{1p} + \cdots + a_{1n}\,b_{mp} \\ \vdots & \vdots & \vdots \\ a_{n1}\,b_{11} + \cdots + a_{nm}\,b_{n1} & \cdots & a_{n1}\,b_{1p} + \cdots + a_{nm}\,b_{mp} \end{bmatrix}.
\end{aligned} \tag{A.30}
$$

If $\mathbf{A}$ is $n \times m$, and $\mathbf{B}$ is $m \times p$, the product $\mathbf{A}\,\mathbf{B}$ is $n \times p$. Additionally

$$
(\mathbf{A}\,\mathbf{B})' = \mathbf{B}'\,\mathbf{A}'. \tag{A.31}
$$

We note in passing that $\mathbf{AB}$ can also be viewed as a linear combination of the rows of $\mathbf{B}$, with the weights being given by the rows of $\mathbf{A}$. In general

$$
\mathbf{A}\,\mathbf{B} \neq \mathbf{B}\,\mathbf{A}. \tag{A.32}
$$

Matrices prove particularly useful when solving sets of linear equations. If we set

$$
\mathbf{A}\,\mathbf{x} = \mathbf{b}, \tag{A.33}
$$

where $\mathbf{b}$ is a column vector, we have a compact representation of a collection of linear equations that can be interpreted in one of two ways:

1. Row perspective: Each row of (A.33) corresponds to a *hyperplane*[8], and solving (A.33) entails finding a vector $\mathbf{x}$ that lies at the intersection of these $n$ hyperplanes
2. Column perspective: Each column of $\mathbf{A}$ is a vector, and solving (A.33) entails finding a vector $\mathbf{x}$ that makes the two sides of (A.33) equal.

Sometimes, (A.33) possesses a unique solution, while at other times, it can have an infinite number of solutions or no solution at all. The existence of a unique solution is governed by whether or not $\mathbf{A}$ is invertible, a topic we will return to shortly.

[8] In $n$–dimensional Euclidean space, a hyperplane is a subspace whose dimension is one less than the dimension of the space. In 2 and 3 dimensions, hyperplanes are lines and planes respectively.

### *A.2.4 Some Special Matrices*

Some matrices have a special structure that greatly simplifies operations on them[9].

If a matrix is square, and if $a_{ij} = a_{ji}$ for all $i$, $j$, it is said to be *symmetric*. If, in addition, $a_{ij} = 0 \; \forall i \neq j$, it is said to be *diagonal*.The transpose of a symmetric matrix is equal to itself. Clearly, all diagonal matrices are symmetric, but the converse is not true.

The *Zero matrix* $\mathbf{0}_n$ is an $n \times n$ matrix with 0's everywhere, i.e.

$$\mathbf{0}_n = \begin{bmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}. \tag{A.34}$$

The *Identity matrix* $\mathbf{I}_n$ is an $n \times n$ diagonal matrix with 1's along the diagonal and 0's elsewhere, i.e.

$$\mathbf{I}_n = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}. \tag{A.35}$$

$\mathbf{J}_n$, the $n \times n$ matrix with all 1's can be constructed by mutliplying a vector of $n$ 1's by its transpose

$$\begin{aligned} \mathbf{J}_n &= \mathbf{1}_n \mathbf{1}_n' \\ &= \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}' \\ &= \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix}. \end{aligned} \tag{A.36}$$

An $n \times m$ matrix is said to be *Orthogonal* if

$$\mathbf{A}' \mathbf{A} = \mathbf{I}_m, \tag{A.37}$$

and it is useful to think of such a matrix as a stacked collection of vectors that lie at right angles to each other in Euclidean space. The product $\mathbf{A}'\mathbf{A}$ is a symmetric matrix that is known as the *Gramian*, and whose rank is the same as the rank of $\mathbf{A}$.

A martrix $\mathbf{A}$ can be raised to an integer power $k$ by multiplying together $k$ copies of itself. For example

$$\mathbf{A}^2 = \mathbf{A}\mathbf{A}, \tag{A.38}$$

and is said to be *nilpotent of index k* if

[9] A comprehensive list of matrices with special structures and their names can be found at `https://en.wikipedia.org/wiki/List_of_named_matrices`.

$$\mathbf{A}^k = \mathbf{0}_n \tag{A.39a}$$

$$\mathbf{A}^j \neq \mathbf{0}_n \text{ for } 1 \leqslant j < k. \tag{A.39b}$$

### *A.2.5 Toeplitz Matrices*

A *Toeplitz* matrix is a square matrix with constant offset diagonals, i.e.

$$a_{i,j} = a_{i+1,j+1},\ 1 \leqslant i, j \leqslant n-1 \tag{A.40}$$

e.g.,

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 1 & 2 & 3 & 4 \\ 7 & 6 & 1 & 2 & 3 \\ 8 & 7 & 6 & 1 & 2 \\ 9 & 8 & 7 & 6 & 1 \end{bmatrix}. \tag{A.41}$$

In the special case when each row is a cyclic shift of the prior row, the matrix is referred to as a *circulant* matrix, e.g.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{bmatrix}. \tag{A.42}$$

Every circulant matrix is a Toeplitz matrix, but the converse is not true.

### *A.2.6 Block Matrices*

A matrix can often usefully be partitioned into a set of adjacent *blocks,* with the boundaries between blocks being identified by vertical and horizontal lines. The circulant matrix displayed in (A.42), for example, can be partitioned into four blocks in a variety of ways as can be seen in (A.43a)–(A.43d). Observe that the various submatrices are not necessarily square – in particular, *none* of the submatrices in (A.43d) are square.

Even though a matrix can have multiple blocks, we illustrate block matrix operations using matrices that are partitioned into four adjacent blocks. The results presented here generalize naturally to larger numbers of blocks, and we refer the interested reader to the comprehensive treatment in Gentle (2017).

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{bmatrix} \tag{A.43a}$$

$$= \left[\begin{array}{c|c} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \hline \mathbf{A}_{21} & \mathbf{A}_{22} \end{array}\right] \tag{A.43b}$$

$$= \left[\begin{array}{cc|ccc} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \\ \hline 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{array}\right] \tag{A.43c}$$

$$= \left[\begin{array}{ccc|cc} 1 & 2 & 3 & 4 & 5 \\ \hline 5 & 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{array}\right] \tag{A.43d}$$

It is useful to think of the blocks along the main diagonal as representing the influence of a tightly coupled set of items on each other (e.g. stocks within a sector, with one block per sector), while blocks off the main diagonal represent interactions between different sets of items (i.e. between sectors in an index). Sometimes, the off–diagonal blocks have entries that are much smaller in magnitude than the blocks on the diagonal, and may be set to 0 to simplify the matrix. A matrix whose off–diagonal blocks are all zero is called a *block diagonal* matrix.

The transpose of a block matrix is obtained by block transposition of the transposes of each of the blocks:

$$\begin{aligned} \mathbf{A}' &= \left[\begin{array}{c|c} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \hline \mathbf{A}_{21} & \mathbf{A}_{22} \end{array}\right]' \\ &= \left[\begin{array}{c|c} \mathbf{A}'_{11} & \mathbf{A}'_{21} \\ \hline \mathbf{A}'_{12} & \mathbf{A}'_{22} \end{array}\right]. \end{aligned} \tag{A.44}$$

Just as matrices allow addition and multiplication if their dimensions are conformable, so do block matrices allow block addition and block multiplication provided that the dimensions of the various blocks are conformable, e.g.

$$\begin{aligned} \mathbf{A} + \mathbf{B} &= \left[\begin{array}{c|c} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \hline \mathbf{A}_{21} & \mathbf{A}_{22} \end{array}\right] + \left[\begin{array}{c|c} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \hline \mathbf{B}_{21} & \mathbf{B}_{22} \end{array}\right] \\ &= \left[\begin{array}{c|c} \mathbf{A}_{11} + \mathbf{B}_{11} & \mathbf{A}_{12} + \mathbf{B}_{12} \\ \hline \mathbf{A}_{21} + \mathbf{B}_{21} & \mathbf{A}_{22} + \mathbf{B}_{22} \end{array}\right], \end{aligned} \tag{A.45}$$

and

$$\mathbf{A}\,\mathbf{B} = \left[\begin{array}{c|c} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \hline \mathbf{A}_{21} & \mathbf{A}_{22} \end{array}\right] \left[\begin{array}{c|c} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \hline \mathbf{B}_{21} & \mathbf{B}_{22} \end{array}\right]$$

$$= \left[\begin{array}{c|c} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \hline \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{array}\right]. \tag{A.46}$$

The partitioned matrix in (A.43c) can be multiplied by itself in accordance with (A.46), but the partition in (A.43d) cannot be block multiplied by itself or by the partition in (A.43c).

### *A.2.7 Echelon Form and LU Factorization*

Computing the rank of a matrix has much in common with solving the set of linear equations specified in (A.33). Using Gaussian elimination[10], one can transform $\mathbf{A}$ into a matrix $\mathbf{E}$ in *row echelon form* with the following properties

1. All non-zero rows of $\mathbf{E}$ lie above all rows with only zeros
2. The first non–zero entry in each row lies strictly to the right of the first non–zero entry in the row above it, and
3. All entries below the first non–zero entry in each row are 0.

The rank of $\mathbf{A}$ is the number of non–zero rows in $\mathbf{E}$. It is also the dimension of its column and row spaces. The sum of the dimensions of the column space and the null space must equal the number of columns, while the sum of the dimensions of the row space and the left null space must equal the number of rows.

If $\mathbf{A}$ is square and of full rank, it can be represented as the product of lower and upper triangular matrices $\mathbf{L}$ and $\mathbf{U}$, i.e.

$$\mathbf{A} = \mathbf{L}\,\mathbf{U}, \tag{A.47}$$

where $\mathbf{L}$ has only 1's on the diagonal, permitting a rapid solution to (A.33). A particular useful form of the LU factorization is the LDU factorization

$$\mathbf{A} = \mathbf{L}\,\mathbf{D}\,\mathbf{U}, \tag{A.48}$$

where $\mathbf{D}$ is a diagonal matrix, and both $\mathbf{L}$ and $\mathbf{U}$ have only 1's on the diagonal.

### *A.2.8 The Determinant of a Matrix*

The determinant of a square matrix $\mathbf{A}$, denoted $det\,(\mathbf{A})$, is the signed volume of a parallelepiped whose vertices are the $2^n$ possible sums of the rows of $\mathbf{A}$[11]. Unlike the volume of a physical object, which

[10] Gaussian elimination attempts to triangularize a matrix by iteratively subtracting a multiple of each row, starting from the first row and working downwards, from all the rows below so as to cancel their leading entry. The process transforms a matrix into row echelon form, and ideally into an upper triangular matrix.

[11] The origin, which corresponds to the sum of none of the rows, is one of the vertices of the parallelepiped.

is always positive, the determinant of a matrix has both sign and magnitude. In this regard, it has something in common with vectors. If $det\,(\mathbf{A}) = 0$, $\mathbf{A}$ is said to be *singular*.

It is particularly easy to compute the determinant of a diagonal matrix $\mathbf{D}$, as its $2^n$ possible row sums represent an $n-$dimensional hypercube with orthogonal sides. Its volume (and determinant) are therefore given by

$$det\,(\mathbf{D}) = \prod_{i=1}^{n} d_{ii}. \tag{A.49}$$

The row sums of the $n \times n$ identity matrix $\mathbf{I}_n$ represent a unit hypercube, and its determinant must therefore be 1.

More generally, the determinant of a block diagonal matrix is the product of the determinants of each of its diagonal blocks, i.e.

$$det \begin{bmatrix} \mathbf{D}_{11} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \ddots & & \mathbf{0} \\ \vdots & & & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{D}_{nn} \end{bmatrix} = \prod_{i=1}^{n} det\,(\mathbf{D}_{ii})\,. \tag{A.50}$$

A $2 \times 2$ matrix vividly illustrates many properties of determinants, though it is not always possible to project the insights it affords to understand larger matrices. It is easy to show that the area of a quadrilateral with vertices $(0, 0)$, $(a, b)$, $(c, d)$ and $(a+c, b+d)$ is $ad - bc$, and it follows that

$$det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc. \tag{A.51}$$

If any two rows of $\mathbf{A}$ are interchanged, the magnitude of its determinant remains unchanged, but its sign reverses – indicating that we are visiting the vertices of the hypercube in a different sequence, e.g.

$$\begin{aligned} det \begin{bmatrix} c & d \\ a & b \end{bmatrix} &= bc - ad \\ &= -det \begin{bmatrix} a & b \\ c & d \end{bmatrix}. \end{aligned} \tag{A.52}$$

Furthermore, if any two rows of $\mathbf{A}$ are identical, its determinant must be 0. This follows from the observation that exchanging two rows must reverse the sign of the determinant. But if the rows are identical, the determinant must remain the same, and this can only happen if it is 0.

Multiplying any single row of $\mathbf{A}$ by a constant $c$ scales its determinant by $c$ as well – the volume of a hypercube is proportional to the length of each of its sides. It follows that if the entire matrix is multiplied by $c$, its determinant is scaled by $c^n$, i.e.

$$det\,(c\mathbf{A}) = c^n det\,(\mathbf{A})\,. \tag{A.53}$$

If any row of $\mathbf{A}$ is 0, so is its determinant, as this is equivalent to multiplying that row by 0.

Matrix transposition does not impact the value of the determinant, i.e.

$$det\left(\mathbf{A}'\right) = det\left(\mathbf{A}\right), \tag{A.54}$$

and by way of example, it is easy to check that the area of a quadrilateral with vertices $(0,0)$, $(a,d)$, $(b,c)$ and $(a+b, c+d)$ is $ad-bc$. This is *not* the same quadrilateral as that associated with (A.51), but their areas are identical.

Additionally, adding a multiple of any row to any other row (or any column to any other column) does not change the value of the determinant, e.g.

$$\begin{aligned} det\begin{bmatrix} a & b \\ c+na & d+nb \end{bmatrix} &= a\left(d+nb\right) - b\left(c+na\right) \\ &= ad - bc \\ &= det\begin{bmatrix} a & b+ma \\ c & d+mc \end{bmatrix} \\ &= det\begin{bmatrix} a & b \\ c & d \end{bmatrix}. \end{aligned} \tag{A.55}$$

In particular, if $a \neq 0$ and $n = -c/a$ in (A.55), the matrix becomes upper triangular, and its determinant is just the product of its diagonal entries.

$$det\begin{bmatrix} a & b \\ 0 & \frac{ad-bc}{a} \end{bmatrix} = ad - bc. \tag{A.56}$$

More generally,

$$\begin{aligned} det\begin{bmatrix} t_{11} & t_{12} & \cdots & t_{1n} \\ 0 & t_{22} & & t_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & t_{nn} \end{bmatrix} &= det\begin{bmatrix} t_{11} & 0 & \cdots & 0 \\ t_{21} & t_{22} & & 0 \\ \vdots & \vdots & \ddots & \vdots \\ t_{n1} & t_{n2} & \cdots & t_{nn} \end{bmatrix} \\ &= \prod_{i=1}^{n} t_{ii}. \end{aligned} \tag{A.57}$$

This follows from the observation that elementary row operations can be used to to combine rows and diagonalize a triangular matrix without changing its determinant. In a like manner, the determinant of a block triangular matrix is the product of the determinants of its diagonal blocks:

$$det \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} & \cdots & \mathbf{T}_{1n} \\ \mathbf{0} & \mathbf{T}_{22} & & \mathbf{T}_{2n} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{T}_{nn} \end{bmatrix} = det \begin{bmatrix} \mathbf{T}_{11} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{T}_{21} & \mathbf{T}_{22} & & \mathbf{0} \\ \vdots & & \ddots & \vdots \\ \mathbf{T}_{n1} & \mathbf{T}_{n2} & \cdots & \mathbf{T}_{nn} \end{bmatrix}$$

$$= \prod_{i=1}^{n} det\left(\mathbf{T}_{ii}\right). \tag{A.58}$$

Determinants are multiplicative:

$$det\left(\mathbf{AB}\right) = det\left(\mathbf{A}\right) det\left(\mathbf{B}\right), \tag{A.59}$$

and, as a consequence, the LDU factorization of a matrix, which is of great practical importance, preserves the value of its determinant. The determinants of all three matrices are given by the product of their diagonal entries, and the determinants of both **L** and **U** are 1, so that the determinant of **D** must be identical to that of **A**.

### *A.2.9 The Inverse of a Matrix*

If an $n \times n$ square matrix **A** is of full rank, its determinant is non–zero, and it is said to *invertible*; there exists a unique $n \times n$ square matrix, denoted $\mathbf{A}^{-1}$, such that

$$\begin{aligned} \mathbf{A}^{-1}\mathbf{A} &= \mathbf{A}\mathbf{A}^{-1} \\ &= \mathbf{I}_n. \end{aligned} \tag{A.60}$$

Matrix inversion can be thought of as the generalization of the reciprocal to matrices, and this interpretation is clearly visible when inverting a diagonal matrix . If

$$\mathbf{D} = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix}$$

$$\mathbf{D}^{-1} = \begin{bmatrix} \frac{1}{d_{11}} & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \frac{1}{d_{nn}} \end{bmatrix}$$

$$= \frac{1}{det\left(\mathbf{D}\right)} \begin{bmatrix} \prod_{i \neq 1} d_{ii} & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \prod_{i \neq n} d_{ii} \end{bmatrix}. \tag{A.61}$$

In general, the inverse of a matrix is given by the ratio of the transpose of its *cofactor matrix* [12] and its determinant

$$\mathbf{A}^{-1} = \frac{\mathbf{F}'}{det\,(\mathbf{A})}, \tag{A.62}$$

where $f_{ij}$ is $(-1)^{i+j}$ times the determinant of $\mathbf{A}$ with its $i^{th}$ row and $j^{th}$column removed. The inverse of a $2 \times 2$ matrix, for example, is

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}. \tag{A.63}$$

The inverse of a product of two matrices exhibits the same "reverse order" property we saw earlier for its transpose:

$$(\mathbf{A}\,\mathbf{B})^{-1} = \mathbf{B}^{-1}\,\mathbf{A}^{-1}, \tag{A.64}$$

and the availability of an inverse allows a closed form solution to (A.33):

$$\mathbf{x} = \mathbf{A}^{-1}\,\mathbf{b}. \tag{A.65}$$

Somewhat surprisingly, and in spite of its great conceptual importance, matrix inversion is rarely used when solving linear equations on account of the fact that the the computation of the cofactor matrix is both tedious and numerically unstable. Instead, Gaussian elimination and LU factorization are used to solve (A.33) directly.

### *A.2.10 Inverses of Special Matrices*

There appears to be a virtually inexhaustible supply of results involving the inverses of block matrices of various types – the results we present here are those most likely to prove useful in portfolio construction and risk management and are drawn from Lu and Shiou (2002). $2 \times 2$ block matrices prove particularly useful, as they are often used to split securities into two subsets (loosely speaking, a set of securities of interest and another set containing all other securities). Formulas involving the inverse of a submatrix are valid only if the inverse exists. Likewise, formulas that involve the product of matrices are valid only if the relevant submatrices are conformable.

We start by stating results for the inverses of diagonal and anti–diagonal block matrices[13], and indicate their sizes via subscripts:

$$\begin{bmatrix} \mathbf{A}_{n\times n} & \mathbf{0}_{n\times m} \\ \mathbf{0}_{m\times n} & \mathbf{B}_{m\times m} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{A}^{-1}_{n\times n} & \mathbf{0}_{n\times m} \\ \mathbf{0}_{m\times n} & \mathbf{B}^{-1}_{m\times m} \end{bmatrix} \tag{A.66a}$$

$$\begin{bmatrix} \mathbf{0}_{n\times m} & \mathbf{A}_{n\times n} \\ \mathbf{B}_{m\times m} & \mathbf{0}_{m\times n} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{0}_{m\times n} & \mathbf{B}^{-1}_{m\times m} \\ \mathbf{A}^{-1}_{n\times n} & \mathbf{0}_{n\times m} \end{bmatrix}. \tag{A.66b}$$

[12] We use the letter $\mathbf{F}$ to denote the cofactor matrix as $\mathbf{C}$ is used to denote the covariance matrix.

[13] We take the liberty of naming the blocks $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ and not $\mathbf{A}_{11}$, $\mathbf{A}_{12}$, $\mathbf{A}_{21}$ and $\mathbf{A}_{22}$ solely for clarity of exposition.

We next provide expressions for the inverses of upper and lower triangular matrices that are valid if $\mathbf{A}$ and $\mathbf{C}$ are invertible:

$$\begin{bmatrix} \mathbf{A}_{n\times n} & \mathbf{B}_{n\times m} \\ \mathbf{0}_{m\times n} & \mathbf{C}_{m\times m} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0}_{m\times n} & \mathbf{C}^{-1} \end{bmatrix} \tag{A.67a}$$

$$\begin{bmatrix} \mathbf{A}_{n\times n} & \mathbf{0}_{n\times m} \\ \mathbf{B}_{m\times n} & \mathbf{C}_{m\times m} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{A}^{-1} & \mathbf{0}_{n\times m} \\ -\mathbf{C}^{-1}\mathbf{B}\mathbf{A}^{-1} & \mathbf{C}^{-1} \end{bmatrix}. \tag{A.67b}$$

In the important special case when $\mathbf{A} = \mathbf{I}_n$ and $\mathbf{C} = \mathbf{I}_m$, (A.67a) and (A.67b) simplify to

$$\begin{bmatrix} \mathbf{I}_n & \mathbf{B}_{n\times m} \\ \mathbf{0}_{m\times n} & \mathbf{I}_m \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{I}_n & -\mathbf{B}_{n\times m} \\ \mathbf{0}_{m\times n} & \mathbf{I}_m \end{bmatrix} \tag{A.68a}$$

$$\begin{bmatrix} \mathbf{I}_n & \mathbf{0}_{n\times m} \\ \mathbf{B}_{m\times n} & \mathbf{I}_m \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{I}_n & \mathbf{0}_{n\times m} \\ -\mathbf{B}_{m\times n} & \mathbf{I}_m \end{bmatrix}. \tag{A.68b}$$

In the general case when all 4 matrices are non-zero, the inverse is obtained by inverting each of the matrices in a block LDU or UDL representations using (A.66a), (A.68a) and (A.68b), and finally combining their inverses using (A.64):

$$\begin{bmatrix} \mathbf{A}_{n\times n} & \mathbf{B}_{n\times m} \\ \mathbf{C}_{m\times n} & \mathbf{D}_{m\times m} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_n & \mathbf{0}_{n\times m} \\ \mathbf{C}\mathbf{A}^{-1} & \mathbf{I}_m \end{bmatrix} \begin{bmatrix} \mathbf{A} & \mathbf{0}_{n\times m} \\ \mathbf{0}_{m\times n} & \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B} \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{0}_{m\times n} & \mathbf{I}_m \end{bmatrix} \tag{A.69a}$$

$$= \begin{bmatrix} \mathbf{I}_n & \mathbf{B}\mathbf{D}^{-1} \\ \mathbf{0}_{m\times n} & \mathbf{I}_m \end{bmatrix} \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C} & \mathbf{0}_{n\times m} \\ \mathbf{0}_{m\times n} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & \mathbf{0}_{n\times m} \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_m \end{bmatrix}. \tag{A.69b}$$

The choice of which expansion to use will depend on whether $\mathbf{A}$ or $\mathbf{D}$ is invertible. Furthermore, if the matrix is to be inverted, either $\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$ or $\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}$ must additionally be invertible.

While formulae are known for the inverse of Toeplitz matrices (see Lv and Huang (2007)), they are cumbersome. In general, the inverse of a Toeplitz matrix is not necessarily a Toeplitz matrix, but the inverse of a circulant matrix is also a circulant matrix. One Toeplitz matrix that sees extensive use in time series work and its inverse are

$$\begin{bmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{n-1} \\ \rho & 1 & \rho & \cdots & \rho^{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{n-2} & \rho^{n-3} & \rho^{n-4} & \cdots & 1 \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \cdots & \rho \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -\rho & 0 & 0 & \cdots & 0 \\ -\rho & 1+\rho^2 & -\rho & 0 & \cdots & 0 \\ 0 & -\rho & 1+\rho^2 & -\rho & \vdots & \vdots \\ \vdots & \vdots & -\rho & 1+\rho^2 & -\rho & 0 \\ 0 & 0 & \cdots & -\rho & 1+\rho^2 & -\rho \\ 0 & 0 & \cdots & 0 & -\rho & 1 \end{bmatrix}. \tag{A.70}$$

### *A.2.11 The Sherman–Morrison–Woodbury Matrix Inversion Lemma*

The following identities see wide application in factor models, as well as in statistics and signal processing. They are variously known as the Sherman–Morrison–Woodbury matrix inversion lemma, the Sherman–Morrison formula, and the Woodbury matrix identity. The fact that they are attributed to many authors is likely a reflection of the fact that they were discovered independently by a multitude of researchers at about the same time.

Broadly speaking, they provide an affirmative answer to the following question: If, after computing the inverse of $\mathbf{A}$, we perturb it with a low rank (as opposed to a low magnitude) perturbation, can we efficiently compute the inverse of the perturbed matrix by perturbing $\mathbf{A}^{-1}$ with a related low rank perturbation[14]?

We start with the special case when the perturbation has rank 1.

Let $\mathbf{A}$ be an invertible $n \times n$ matrix, let $\mathbf{x}$ and $\mathbf{y}$ be column vectors of length $n$ (and hence of rank 1), and let $c$ be any scalar. Then:

$$(\mathbf{A} + c\mathbf{x}\mathbf{y}')^{-1} = \mathbf{A}^{-1} - \frac{c\mathbf{A}^{-1}\mathbf{x}\mathbf{y}'\mathbf{A}^{-1}}{1 + c\mathbf{x}'\mathbf{A}^{-1}\mathbf{y}}. \tag{A.71}$$

Equation (A.71) simplifies even further when $\mathbf{A} = \mathbf{I}_n$, for in this case, $\mathbf{A}^{-1} = \mathbf{A}$, and we get

$$(\mathbf{I}_n + c\mathbf{x}\mathbf{y}')^{-1} = \mathbf{I}_n - \frac{c\mathbf{x}\mathbf{y}'}{1 + c\mathbf{x}'\mathbf{y}}. \tag{A.72}$$

Both (A.71) and (A.72) can be evaluated in $O\left(n^2\right)$ arithmetic operations (direct matrix inversion using Gaussian elimination requires $O\left(n^3\right)$ operations), and even more importantly, they are numerically stable – small changes to any of the parameters on the left hand sides of (A.71) and (A.72) result in small changes to their right hand sides.

More generally, let $\mathbf{A}$ be $n \times n$, $\mathbf{U}$ be $n \times k$, $\mathbf{C}$ be $k \times k$, and $\mathbf{V}$ be $k \times n$. Then

$$(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\left(\mathbf{C}^{-1} + \mathbf{V}\mathbf{A}^{-1}\mathbf{U}\right)^{-1}\mathbf{V}\mathbf{A}^{-1} \tag{A.73}$$

where it is assumed that the inverses all exist. Typically, $k \ll n$, making the inversion of $\mathbf{C}$ far less tedious than direct recomputation of $\mathbf{A}^{-1}$.

Henderson and Searle (1981) and Hager (1989) provide a useful compendium of results related to (A.71), (A.72), and (A.73), as well as a comprehensive summary of how they are applied in a variety of settings. One such result is

[14] Proofs of these, and other related, identities can be found at `https://en.wikipedia.org/wiki/Woodbury_matrix_identity`.

$$\begin{bmatrix} a & b & \cdots & b \\ b & a & \cdots & b \\ \vdots & \vdots & \ddots & \vdots \\ b & b & \cdots & a \end{bmatrix}^{-1} = \left[(a-b)\,\mathbf{I}_n + b\,\mathbf{1}_n\mathbf{1}_n'\right]^{-1}$$

$$= \frac{1}{a-b}\left[\mathbf{I}_n - \frac{b}{a+(n-1)\,b}\mathbf{J}_n\right]. \tag{A.74}$$

where $\mathbf{J}_n = \mathbf{1}_n\mathbf{1}_n'$ is defined in (A.36), and provided that $a \neq b$ and $a + (n-1)\,b \neq 0$.

This result proves particularly useful when analyzing constant correlation matrices with $a = 1$ and $b = \rho$ for $-1/(n-1) < \rho < 1$[15], i.e.

$$\begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix}^{-1} = \frac{1}{1-\rho}\left[\mathbf{I}_n - \frac{\rho}{1+(n-1)\,\rho}\mathbf{J}_n\right]$$

$$= \frac{(1+(n-1)\,\rho)\,\mathbf{I}_n - \rho\,\mathbf{J}_n}{(1-\rho)\,(1+(n-1)\,\rho)}. \tag{A.75}$$

### *A.2.12 The Inverse of a Covariance Matrix*

Sometime, a closed form inverse of a matrix yields useful insights into the nature of its entries, and Stevens (1998) provides a particularly good example of this. Assume that we have a set of $n$ securities. The covariance matrix of their returns can be partitioned as

$$\mathbf{C}_n = \left[\begin{array}{c|c} \sigma_1^2 & \mathbf{\Sigma}_{12} \\ \hline \mathbf{\Sigma}_{21} & \mathbf{C}_{n-1} \end{array}\right], \tag{A.76}$$

where $\sigma_1^2$ is the variance of security 1, $\mathbf{\Sigma}_{12}$ is a $1 \times (n-1)$ vector that contains the covariances of security 1 with the remaining $n-1$ securities, $\mathbf{\Sigma}_{21} = \mathbf{\Sigma}_{12}'$, and $\mathbf{C}_{n-1}$ is the covariance matrix of the remaining $n-1$ securities. We can partition $\mathbf{C}_n^{-1}$ in the same way we partition $\mathbf{C}_n$:

$$\mathbf{C}_n^{-1} = \left[\begin{array}{c|c} a_{11} & \mathbf{\Lambda}_{12} \\ \hline \mathbf{\Lambda}_{21} & \mathbf{\Lambda}_{22} \end{array}\right]. \tag{A.77}$$

As

$$\mathbf{C}_n^{-1}\mathbf{C}_n = \mathbf{I}_n,$$

$$= \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{I}_{n-1} \end{array}\right] \tag{A.78}$$

[15] Outside this range, the matrix may be invertible, but it is not positive definite.

we can write

$$a_{11}\,\sigma_1^2 + \mathbf{\Lambda}_{12}\,\mathbf{\Sigma}_{21} = 1 \tag{A.79a}$$

$$a_{11}\,\mathbf{\Sigma}_{12} + \mathbf{\Lambda}_{12}\,\mathbf{C}_{n-1} = \mathbf{0}. \tag{A.79b}$$

On solving (A.79b) for $\mathbf{\Lambda}_{12}$, substituting into (A.79a) and simplifying, we get

$$\begin{aligned} a_{11} &= \frac{1}{\sigma_1^2 - \mathbf{\Sigma}_{12}\,\mathbf{C}_{n-1}^{-1}\,\mathbf{\Sigma}_{21}} \\ &= \frac{1}{\sigma_1^2\,\left(1 - R_1^2\right)} \\ &= \frac{1}{\sigma_\varepsilon^2}, \end{aligned} \tag{A.80}$$

where $R_1^2$ is the fraction of the variance of security 1 that is explained by securities $2, 3, \cdots, n$ when its returns are regressed against theirs, and $\sigma_\varepsilon^2$ is the variance of the regression residuals. These are well known results in the theory of least squares, and are discussed in Stevens (1997) and Hayashi (2000). Next, we write

$$\begin{aligned} \mathbf{\Lambda}_{12} &= -a_{11}\,\mathbf{\Sigma}_{12}\,\mathbf{C}_{n-1}^{-1} \\ &= \frac{-\mathbf{\Sigma}_{12}\,\mathbf{C}_{n-1}^{-1}}{\sigma_1^2\,\left(1 - R_1^2\right)} \\ &= \frac{-\boldsymbol{\beta}_1}{\sigma_1^2\,\left(1 - R_1^2\right)}, \end{aligned} \tag{A.81}$$

where $\boldsymbol{\beta}_1$ is the vector of regression coefficients obtained when security 1's returns are regressed against those of securities $1, \cdots, i-1, i+1, \cdots, n$.

But since our choice of security 1 is arbitrary, all the rows must have exactly the same form, and the inverse of the covariance matrix must be

$$\mathbf{C}_n^{-1} = \begin{bmatrix} \frac{1}{\sigma_1^2\left(1-R_1^2\right)} & \frac{-\beta_{12}}{\sigma_1^2\left(1-R_1^2\right)} & \cdots & \frac{-\beta_{1n}}{\sigma_1^2\left(1-R_1^2\right)} \\ \frac{-\beta_{21}}{\sigma_2^2\left(1-R_2^2\right)} & \frac{1}{\sigma_2^2\left(1-R_2^2\right)} & & \frac{-\beta_{2n}}{\sigma_2^2\left(1-R_2^2\right)} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{-\beta_{n1}}{\sigma_n^2\left(1-R_n^2\right)} & \frac{-\beta_{n2}}{\sigma_n^2\left(1-R_n^2\right)} & & \frac{1}{\sigma_n^2\left(1-R_n^2\right)} \end{bmatrix}. \tag{A.82}$$

Even though we would not ever invert a covariance matrix in this way (both the data requirements and the computational effort are huge, and the procedure is not numerically stable), the insights it affords are valuable, and helps clarify our thinking about the structure of the inverse of a covariance matrix.

### *A.2.13 The Mahanalobis Distance*

It sometimes proves useful to measure the distance between two random vectors that are drawn from the same *joint* distribution, but whose *marginal* distributions are very different. For example, vectors $\mathbf{u}$ and $\mathbf{v}$ may contain the returns of all the stocks in the S&P 500 on two adjacent days, but the marginal distribution of their first entries (a volatile small cap biotech firm) may be very different than that of their last entry (a steadily growing consumer goods company). In such cases, it proves useful to first express the distance between each component of the two vectors in standard deviation units, and to then compute a standardized distance between them using a variant of (A.11). In the special case when all the components of $\mathbf{u}$ and $\mathbf{v}$ are independent, this gives us

$$d_{\mathbf{u},\mathbf{v}} = \sqrt{\left(\frac{u_1 - v_1}{\sigma_1}\right)^2 + \cdots + \left(\frac{u_n - v_n}{\sigma_n}\right)^2}. \tag{A.83}$$

More generally, the *Mahalanobis distance* takes into account the covariances between the components of the vectors, and is defined as

$$d^M_{\mathbf{u},\mathbf{v}} = \sqrt{(\mathbf{u}-\mathbf{v})'\,\mathbf{C}^{-1}\,(\mathbf{u}-\mathbf{v})}, \tag{A.84}$$

where $C$ is the covariance matrix of the components of $\mathbf{u}$ (and $\mathbf{v}$). If the covariance matrix is diagonal (i.e. if all the components of a vector are uncorrelated with each other), (A.84) reduces to (A.83).

### *A.2.14 Eigenvalues and Eigenvectors*

In (A.33), a matrix $\mathbf{A}$ maps a non–zero vector $\mathbf{x}$ to another vector $\mathbf{b}$ which may lie at a great distance from $\mathbf{x}$. But not all matrices push vectors far away: the identity matrix $\mathbf{I}_n$, for example, maps every vector to itself, and a scalar multiple of $\mathbf{I}_n$ maps a vector to the same scalar multiple of itself. This property is not unique to $\mathbf{I}_n$ – all square matrices map certain special vectors to scaled versions of themselves, i.e

$$\mathbf{A}\,\mathbf{e} = \lambda\,\mathbf{e}. \tag{A.85}$$

The vector $\mathbf{e}$ is known as an *eigenvector* of $\mathbf{A}$, the scalar $\lambda$ is its corresponding *eigenvalue*, and they are collectively referred to as an *eigenpair*. Note that $\mathbf{e}$ is not unique – any scalar multiple of $\mathbf{e}$ is also an eigenvector of $\mathbf{A}$. Often, an eigenvector is scaled so that its norm is 1, and the resulting eigenvector is known as a *unit eigenvector* or a *normalized eigenvector*.

If we rewrite (A.85) as

$$(\mathbf{A} - \lambda\,\mathbf{I})\,\mathbf{e} = \mathbf{0}, \tag{A.86}$$

the eigenvalues of $\mathbf{A}$ can be seen to be those values of $\lambda$ for which $\mathbf{A} - \lambda\mathbf{I}$ is singular, or non–invertible, and therefore has a null space, or *eigenspace*; any vector $\mathbf{e}$ that lies in the eigenspace associated with a particular eigenvalue is an eigenvector of $\mathbf{A}$. It follows from the definition of invertibility that

$$det\,(\mathbf{A} - \lambda\,\mathbf{I}) = 0, \tag{A.87}$$

Let $\lambda_i, 1 \leqslant i \leqslant n$ be an eigenvalue of an $n \times n$ matrix $\mathbf{A}$. Then

$$\begin{aligned}\sum_{i=1}^{n} \lambda_i &= \sum_{i=1}^{n} a_{ii} \\ &\triangleq trace\left(\mathbf{A}\right),\end{aligned} \tag{A.88}$$

and

$$\prod_{i=1}^{n} \lambda_i = det\left(\mathbf{A}\right). \tag{A.89}$$

The set of eigenvalues of a matrix $\mathbf{A}$ is known as its *spectrum*, and its spectral radius, $\rho\left(\mathbf{A}\right)$, is defined to be

$$\rho\left(\mathbf{A}\right) = \max_i \left|\lambda_i\right|, \tag{A.90}$$

and it can be shown that

$$\rho\left(\mathbf{A}\right) \leqslant \min\left(\max_i \sum_{j=1}^{n} \left|a_{ij}\right|,\ \max_j \sum_{i=1}^{n} \left|a_{ij}\right|\right). \tag{A.91}$$

A few useful facts about the eigenpairs $\lambda_i$ and $\mathbf{e}_i$ of an $n \times n$ matrix $\mathbf{A}$ follow:

1. If $\mathbf{A}$ is invertible, $1/\lambda_i$ and $\mathbf{e}_i$ are an eigenpair of $\mathbf{A}^{-1}$.
2. The eigenvalues (but not necessarily the eigenvectors) of $\mathbf{A}'$ are identical to the eigenvalues of $\mathbf{A}$.
3. $\lambda_i^k$ and $\mathbf{e}_i$ are an eigenpair of $\mathbf{A}^k,\ k > 1$.
4. If $\mathbf{A}$ is triangular or diagonal, then $\lambda_i = a_{ii},\ 1 \leqslant i \leqslant n$ .
5. $\lambda_i - k$ and $\mathbf{e}_i$ are an eigenpair of $\mathbf{A} - k\,\mathbf{I}_n$.
6. If $\mathbf{B}$ is an invertible $n \times n$ matrix, the eigenvalues (but not necessarily the eigenvectors) of $\mathbf{B}^{-1}\,\mathbf{A}\,\mathbf{B}$ are identical to those of $\mathbf{A}$.
7. If the eigenvalues of $\mathbf{A}$ are distinct, or if $\mathbf{A}$ is symmetric and has only real entries (covariance and correlation matrices are good examples of this second, weaker, condition), the matrix $\mathbf{P}$ formed by stacking its normalized (i.e. of length 1) eigenvectors side by side diagonalizes $\mathbf{A}$, i.e.

$$\mathbf{P}^{-1}\,\mathbf{A}\,\mathbf{P} = \mathbf{D}_\lambda \tag{A.92}$$

where

$$\mathbf{D}_\lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix}. \tag{A.93}$$

Equation (A.92) is often written as

$$\mathbf{A} = \mathbf{P}\,\mathbf{D}_\lambda\,\mathbf{P}^{-1}, \tag{A.94}$$

where it is called the *eigendecomposition*, or the *spectral representation*, of $\mathbf{A}$.

It can further be shown that when the eigenvalues of $\mathbf{A}$ are distinct, or if $\mathbf{A}$ is symmetric and has only real entries,

$$\mathbf{P}^{-1} = \mathbf{P}',$$

so that (A.94) can usefully be rewritten as

$$\begin{aligned}\mathbf{A} &= \mathbf{P}\,\mathbf{D}_\lambda\,\mathbf{P}' \\ &= \sum_{i=1}^{n} \lambda_i\,\mathbf{p}_i\,\mathbf{p}_i',\end{aligned} \tag{A.95}$$

where $\mathbf{p}_i$ is the $i^{th}$ column of $\mathbf{P}$.

Clearly, $\mathbf{A}$ is invertible only if $\mathbf{D}_\lambda$ is invertible, and $\mathbf{D}_\lambda$ in turn is invertible if and only if all its eigenvalues are non–zero. Direct multiplication shows that

$$\begin{aligned}\mathbf{A}^k &= \mathbf{P}\,\mathbf{D}_\lambda^k\,\mathbf{P}^{-1} \\ &= \mathbf{P}\,\mathbf{D}_\lambda^k\,\mathbf{P}' \\ &= \sum_{i=1}^{n} \lambda_i^k\,\mathbf{p}_i\,\mathbf{p}_i',\ k \geqslant 2,\end{aligned} \tag{A.96}$$

under the above conditions, and it is also true that

$$\begin{aligned}\exp(\mathbf{A}) &= \mathbf{P}\,\exp(\mathbf{D}_\lambda)\,\mathbf{P}^{-1} \\ &= \mathbf{P}\,\exp(\mathbf{D}_\lambda)\,\mathbf{P}',\end{aligned} \tag{A.97}$$

where

$$\exp(\mathbf{A}) \triangleq \mathbf{I}_n + \sum_{i>0} \frac{1}{i!}\mathbf{A}^i \tag{A.98}$$

is known as the *matrix exponential*.

It follows that

$$\exp(\mathbf{0}_n) = \mathbf{I}_n. \tag{A.99}$$

In the special case when $\mathbf{A}$ and $\mathbf{B}$ commute, i.e. when $\mathbf{A}\,\mathbf{B} = \mathbf{B}\,\mathbf{A}$,

$$\begin{aligned}\exp(\mathbf{A})\exp(\mathbf{B}) &= \exp(\mathbf{B})\exp(\mathbf{A}) \\ &= \exp(\mathbf{A}+\mathbf{B}).\end{aligned} \tag{A.100}$$

In particular, as $\mathbf{A}$ commutes with itself, and as $\mathbf{A}+(-\mathbf{A}) = \mathbf{0}_n$

$$\exp(\mathbf{A})^{-1} = \exp(-\mathbf{A}). \tag{A.101}$$

Equation (A.92) can be generalized: two matrices, $\mathbf{A}$ and $\mathbf{B}$, are said to be *similar* if there exists an invertible matrix $\mathbf{E}$ such that

$$\mathbf{E}^{-1}\mathbf{A}\mathbf{E} = \mathbf{B}. \tag{A.102}$$

Similar matrices have identical eigenvalues, as

$$\begin{aligned}det(\mathbf{B}-\lambda\mathbf{I}) &= det\left(\mathbf{E}^{-1}(\mathbf{A}-\lambda\mathbf{I})\mathbf{E}\right) \\ &= det\left(\mathbf{E}^{-1}\right)det(\mathbf{A}-\lambda\mathbf{I})\,det(\mathbf{E}) \\ &= det(\mathbf{A}-\lambda\mathbf{I}).\end{aligned} \tag{A.103}$$

### *A.2.15 Positive Definite Matrices*

A matrix $\mathbf{A}$ is said to be *positive definite* if

$$\mathbf{x}'\mathbf{A}\mathbf{x} > \mathbf{0}\ \forall\mathbf{x} \neq \mathbf{0}. \tag{A.104}$$

An equivalent condition for positive definiteness is that all the eigenvalues of $\mathbf{A}$ are strictly positive. If we relax the inequality slightly, i.e. if

$$\mathbf{x}'\mathbf{A}\mathbf{x} \geqslant \mathbf{0}\ \forall\mathbf{x} \neq \mathbf{0}, \tag{A.105}$$

$\mathbf{A}$ is said to be *positive semi–definite*.

Negating each entry in a positive definitive (or positive semi–definite) matrix results in a *negative definite* (or *negative semi–definite*) matrix – the inequalities in (A.104) and (A.105) are reversed.

The Gramian matrix $\mathbf{A}'\mathbf{A}$ is non–negative definite, as

$$\begin{aligned}\mathbf{x}'\mathbf{A}'\mathbf{A}\mathbf{x} &= (\mathbf{A}\mathbf{x})'(\mathbf{A}\mathbf{x}) \\ &\geqslant \mathbf{0}\ \forall\mathbf{x}.\end{aligned} \tag{A.106}$$

Furthermore, the rank of the Gramian is identical to the rank of $\mathbf{A}$.

If $\mathbf{A}$ is symmetric and positive semi–definite, its eigenvalues are non–negative. If it is symmetric and positive definite, its eigenvalues are strictly positive.

Positive definite matrices are of great importance in portfolio management because they are used to compute the variance of a portfolio from the variances and covariances of its constituent securities (the variance of a portfolio is, by definition, non–negative), and also find use when identifying extrema of multivariate functions.

As a positive definite covariance matrix is symmetric and has only real entries, (A.92)–(A.95) apply, and we can write

$$\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{D}_\lambda, \tag{A.107}$$

where $\mathbf{P}$ is an orthonormal matrix whose columns are the normalized eigenvalues of $\mathbf{A}$ and $\mathbf{D}_\lambda$ is a diagonal matrix of eigenvalues – see (A.93). If we define

$$\sqrt{\mathbf{D}_\lambda^{-1}} = \begin{bmatrix} 1/\sqrt{\lambda_1} & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & 1/\sqrt{\lambda_n} \end{bmatrix} \tag{A.108}$$

and

$$\mathbf{Q} = \mathbf{P}\sqrt{\mathbf{D}_\lambda^{-1}}, \tag{A.109}$$

we can write

$$\mathbf{Q}'\mathbf{A}\mathbf{Q} = \mathbf{I}_n. \tag{A.110}$$

Using straightforward properties of Cholesky factorization and products of orthogonal matrices, it can be shown that $\mathbf{Q}$ diagonalizes *any* positive definite covariance matrix, say $\mathbf{B}$, i.e.

$$\mathbf{Q}'\mathbf{B}\mathbf{Q} = \mathbf{D}_1, \tag{A.111}$$

where $\mathbf{D}_1$ is a diagonal matrix, though its entries are not the eigenvalues of $\mathbf{B}$. The matrices $\mathbf{A}$ and $\mathbf{B}$ are said to be simultaneously diagonalizable.

### *A.2.16 QR Decomposition and the Gram–Schmidt Procedure*

Assume that we have a set of vectors $\mathbf{q}_1, \mathbf{q}_{2,} \cdots, \mathbf{q}_n$ which form an orthonormal basis[16] for an $n$–dimensional vector space $\mathcal{S}$ . From the definition of orthonormality,

$$\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases} \tag{A.112}$$

If the $\mathbf{q}_i$ are vectors in $n$–dimensional Euclidean space, we can stack them side by side to form an $n \times n$ matrix

[16] An orthogonal basis can be transformed into an orthonormal basis by dividing each vector by its norm.

$$\mathbf{Q} = [\mathbf{q}_1 \cdots \mathbf{q}_n], \tag{A.113}$$

and it is easily seen that

$$\mathbf{Q}'\mathbf{Q} = \mathbf{I}_n. \tag{A.114}$$

We can post–multiply both sides of (A.114) by $\mathbf{Q}^{-1}$ to obtain

$$\mathbf{Q}^{-1} = \mathbf{Q}', \tag{A.115}$$

but it is important to note that (A.114) (but not (A.115)) holds even if $\mathbf{Q}$ is rectangular, i.e. if the $\mathbf{q}_i$ are vectors in an $m$–dimensional Euclidean space, for $m > n$. Also, as $\mathbf{I}_n' = \mathbf{I}_n$, it follows that

$$\mathbf{Q}'\mathbf{Q} = \mathbf{Q}\mathbf{Q}'. \tag{A.116}$$

Remarkably, *any* matrix whose columns are linearly independent can be decomposed into a product of a orthonormal matrix and an invertible upper triangular matrix[17], i.e.

$$\mathbf{A} = \mathbf{Q}\mathbf{R} \tag{A.117}$$

where $\mathbf{Q}$ is orthonormal and $\mathbf{R}$ is invertible and upper triangular.

The *Gram–Schmidt* procedure is a constructive algorithm for creating $\mathbf{Q}$ from any full rank matrix $\mathbf{A}$.

Let

$$\mathbf{A} = [\mathbf{a}_1 \cdots \mathbf{a}_n]. \tag{A.118}$$

be an $n \times n$ matrix of full rank.

If we divide each column of $\mathbf{A}$ by its norm, we get an equivalent[18] matrix $\mathbf{A}_1$, whose columns all have norm 1.

$$\begin{aligned} \mathbf{A}_1 &= \left[ \frac{\mathbf{a}_1}{\|\mathbf{a}_1\|} \cdots \frac{\mathbf{a}_n}{\|\mathbf{a}_n\|} \right] \\ &\triangleq \left[ \mathbf{a}_1^1 \cdots \mathbf{a}_n^1 \right]. \end{aligned} \tag{A.119}$$

To construct $\mathbf{Q}$ using the Gram–Schmidt procedure, we first set $\mathbf{q}_1 = \mathbf{a}_1^1$, and then iteratively subtract the projections of $\mathbf{a}_i^1$ for $i = 2, 3, \cdots, n$ onto $\mathbf{q}_1, \cdots, \mathbf{q}_{i-1}$ from itself, and finally normalize the resulting vector to create a new vector that is orthonormal to all the preceding vectors. Formally,

[17] Normally, an upper triangular matrix would be represented by $\mathbf{U}$, but it has become the norm to represent it in this particular factorization by $\mathbf{R}$, possibly because all its entries lie to the right of the principal diagonal.

[18] Equivalent in the sense that the column space of $\mathbf{A}_1$ is identical to the column space of $\mathbf{A}$.

$$\begin{aligned}
\mathbf{q}_1 &= \mathbf{a}_1^1 \\
\mathbf{q}_2 &= \frac{\mathbf{a}_2^1 - \langle \mathbf{a}_2^1, \mathbf{q}_1 \rangle \mathbf{q}_1}{\left\| \mathbf{a}_2^1 - \langle \mathbf{a}_2^1, \mathbf{q}_1 \rangle \mathbf{q}_1 \right\|} \\
&\vdots \\
\mathbf{q}_n &= \frac{\mathbf{a}_n^1 - \sum_{i=1}^{n-1} \langle \mathbf{a}_n^1, \mathbf{q}_i \rangle \mathbf{q}_i}{\left\| \mathbf{a}_n^1 - \sum_{i=1}^{n-1} \langle \mathbf{a}_n^1, \mathbf{q}_i \rangle \mathbf{q}_i \right\|}
\end{aligned} \tag{A.120}$$

It is easy to check that the $\mathbf{q}_i$ satisfy (A.112), and that the column space of $\mathbf{Q}$ is identical to the column space of $\mathbf{A}$. Keep in mind that the Gram–Schmidt procedure does not return a unique orthonormal matrix – if, for example, the columns of $\mathbf{A}$ are resequenced, it will return a different orthonormal matrix.

Even though the Gram–Schmidt procedure makes clear the conceptual basis of orthonormalization, it is rarely used in practice on account of its numerically instability when the columns of $\mathbf{A}$ are almost collinear. And though its numerical stability can be improved by resequencing its computations (see Gentle (2017) for details), *Householder transformations* and *Givens rotations* are more widely used in practice to create $\mathbf{QR}$ decompositions, which see widespread use in the solution of (A.33).

The fact that the transpose of an orthonormal matrix is its inverse leads to a subtly different notion of similarity: two matrices, $\mathbf{A}$ and $\mathbf{B}$, are said to be *orthogonally similar* if there exists an orthonormal matrix $\mathbf{F}$ such that

$$\mathbf{F}' \mathbf{A} \mathbf{F} = \mathbf{B}. \tag{A.121}$$

If $\mathbf{B}$ is upper triangular, then $\mathbf{F}\mathbf{B}\mathbf{F}'$is known as the *Schur factorization* of $\mathbf{A}$.

### *A.2.17 The Singular Value Decomposition*

In the special case when $\mathbf{A}$ is square and symmetric, it can be represented as the Gramian of some $n \times m$ matrix $\mathbf{B}$ with rank $r \leqslant \min(m, n)$ (think of the columns of $\mathbf{B}$ as time series of returns of securities, some of which may admit a riskless arbitrage), i.e.,

$$\mathbf{A} = \mathbf{B}' \mathbf{B}. \tag{A.122}$$

The rank of $\mathbf{A}$ is also $r$, and it possesses a set of orthonormal eigenvectors that diagonalize it, i.e. its spectral representation is

$$\mathbf{A} = \mathbf{Q} \mathbf{D}_\lambda \mathbf{Q}', \tag{A.123}$$

where $\mathbf{Q}$ is an orthonormal matrix whose columns are the normalized eigenvectors of $\mathbf{A}$, and $\mathbf{D}_\lambda$ is a diagonal matrix whose entries are the eigenvalues of $\mathbf{A}$.

The *Singular Value Decomposition* (SVD) generalizes and extends (A.123) to *all* matrices, and diagonalizes $\mathbf{A}_{n \times m}$ using its *singular vectors* instead of its eigenvalues. The resulting decomposition represents any $n \times m$ matrix as a product of three matrices

$$\mathbf{A}_{n \times m} = \mathbf{U}_{n \times n} \mathbf{\Sigma}_{n \times m} \mathbf{V}'_{m \times m}, \tag{A.124}$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthonormal matrices, and $\mathbf{\Sigma}$ is diagonal in the following sense:

$$\sigma_{ij} = \begin{cases} \sigma_i > 0 & 1 \leqslant i = j \leqslant \operatorname{rank}(\mathbf{A}) \\ 0 & \text{otherwise.} \end{cases} \tag{A.125}$$

The $\sigma_i$ are known as *singular values*, and are the square roots of the non-zero eigenvalues of $\mathbf{A}'\mathbf{A}$. Correspondingly, the columns of $\mathbf{U}$ are the orthonormal eigenvectors of $\mathbf{A}\mathbf{A}'$ and those of $\mathbf{V}$ are the orthonormal eigenvectors of $\mathbf{A}'\mathbf{A}$. They are known as the *left* and *right singular vectors*, respectively, of $\mathbf{A}$. For symmetric, square, matrices, (A.124) reduces to (A.123).

Recognizing that $\mathbf{V}^{-1} = \mathbf{V}'$,we can rewrite (A.124) as

$$\mathbf{A}_{n\times m}\mathbf{V}_{m\times m} = \mathbf{U}_{n\times n}\mathbf{\Sigma}_{n\times m}, \tag{A.126}$$

so that $\mathbf{A}$ maps the columns of $\mathbf{V}$ to scaled versions of the columns of $\mathbf{U}$. Addtionally, we can partition $\mathbf{U}$ and $\mathbf{V}$ into two adjacent block matrices

$$\mathbf{U} = [\mathbf{U}_1 \mid \mathbf{U}_2] = \left[\mathbf{u}_1 \ \cdots \ \mathbf{u}_r \mid \mathbf{u}_{r+1} \cdots \mathbf{u}_n\right] \tag{A.127a}$$

$$\mathbf{V} = [\mathbf{V}_1 \mid \mathbf{V}_2] = \left[\mathbf{v}_1 \ \cdots \ \mathbf{v}_r \mid \mathbf{v}_{r+1} \cdots \mathbf{v}_m\right], \tag{A.127b}$$

and it can be shown that

$$\begin{aligned}
\mathbf{U}_1 &= \left[\mathbf{u}_1 \ \cdots \ \mathbf{u}_r\right] && \text{is an orthonormal basis for the } \textit{column space} \text{ of } \mathbf{A} \\
\mathbf{U}_2 &= [\mathbf{u}_{r+1} \cdots \mathbf{u}_n] && \text{is an orthonormal basis for } \mathcal{N}(\mathbf{A}'), \text{ the } \textit{left null space} \text{ of } \mathbf{A} \\
\mathbf{V}_1 &= \left[\mathbf{v}_1 \ \cdots \ \mathbf{v}_r\right] && \text{is an orthonormal basis for the } \textit{row space} \text{ of } \mathbf{A} \\
\mathbf{V}_2 &= [\mathbf{v}_{r+1} \cdots \mathbf{v}_m] && \text{is an orthonormal basis for } \mathcal{N}(\mathbf{A}), \text{ the } \textit{null space} \text{ of } \mathbf{A}.
\end{aligned}$$

If we write out (A.124) in terms of the left and right singular vectors $\mathbf{u}_i$ and $\mathbf{v}_i$ contained in $\mathbf{U}_1$ and $\mathbf{V}_1$ (the orthonormal bases for its column space and row space respectively), we see that the SVD allows us to represent$\mathbf{A}$ as a sum of rank 1 matrices:

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i'. \tag{A.128}$$

If we further reorder the terms on the right hand side of (A.128) in descending order of $\sigma_i$, we obtain a decomposition of $\mathbf{A}$ in which the terms are ranked in descending order of importance, and we can obtain a sequence of increasingly accurate approximations to $\mathbf{A}$ by computing the partial sums of the first $p < r$ terms of this reordered sequence.

This is, in fact, the way in which the SVD is most often used – a small, but important, subset of the terms in (A.128) is used to construct an adequate approximation to $\mathbf{A}$ that greatly simplifies computations using it. In many real life applications, matrices with many millions of entries are well approximated by a partial sum with ten or twenty terms, speeding up computations by many orders of magnitude.

## A.3 Some Results on Functions

### *A.3.1 Derivatives of Functions*

Let $f(\mathbf{x})$ be a differentiable function of $n$ variables $x_1, x_2, \cdots, x_n$.
The *gradient* of $f(\mathbf{x})$, $\nabla f(\mathbf{x})$, is a vector of its partial derivatives, and is given by

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix}. \tag{A.129}$$

The gradient vector points in the direction in which $f$'s rate of change is maximal. In the special case when

$$\begin{aligned} f(\mathbf{x}) &= \sum_{i=1}^{n} a_i x_i \\ &= \mathbf{a} \cdot \mathbf{x}, \end{aligned} \tag{A.130}$$

it is easily seen that

$$\nabla f(\mathbf{x}) = \mathbf{a}. \tag{A.131}$$

Next, if $f(\mathbf{x})$ is a *quadratic form*, i.e. if

$$\begin{aligned} f(\mathbf{x}) &= \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j \\ &= \mathbf{x}' \mathbf{A} \mathbf{x}, \end{aligned} \tag{A.132}$$

where the $a_{ij}$ are constants, it can be seen by differentiating and gathering terms that

$$\nabla f(\mathbf{x}) = 2\mathbf{A}\mathbf{x}. \tag{A.133}$$

If $f(\mathbf{x})$ is twice differentiable, its Hessian, $\mathbf{H}(f(\mathbf{x}))$, is a symmetric square matrix of its second partial derivatives:

$$\begin{aligned} \mathbf{H}(f(\mathbf{x})) &= \frac{\partial^2 f(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}'} \\ &= \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n^2} \end{bmatrix} \end{aligned} \tag{A.134}$$

The Hessian matrix reflects the magnitude, as well as the principal axes, of the local curvature of $f(\mathbf{x})$.

If

$$\mathbf{g}(x) = \begin{bmatrix} g_1(x) \\ \vdots \\ g_n(x) \end{bmatrix} \tag{A.135}$$

is a vector of $k$–times differentiable functions, $g_1(x), g_2(x), \cdots, g_n(x)$, of a single variable $x$, then

$$\frac{d^k \mathbf{g}(x)}{dx^k} = \begin{bmatrix} \frac{d^k g_1(x)}{dx^k} \\ \vdots \\ \frac{d^k g_n(x)}{dx^k} \end{bmatrix}. \tag{A.136}$$

Equation (A.136) generalizes naturally to matrices. If

$$\mathbf{G}(x) = \begin{bmatrix} g_{11}(x) & \cdots & g_{1m}(x) \\ \vdots & \vdots & \vdots \\ g_{n1}(x) & \cdots & g_{nm}(x) \end{bmatrix} \tag{A.137}$$

is a matrix of $k$–times differentiable functions, $g_{11}(x), \cdots, g_{nm}(x)$, of a single variable $x$, then

$$\frac{d^k \mathbf{G}(x)}{dx^k} = \begin{bmatrix} \frac{d^k g_{11}(x)}{dx^k} & \cdots & \frac{d g_{1m}^k(x)}{dx^k} \\ \vdots & \vdots & \vdots \\ \frac{d^k g_{n1}(x)}{dx^k} & \cdots & \frac{d^k g_{nm}(x)}{dx^k} \end{bmatrix}. \tag{A.138}$$

### *A.3.2 Taylor Series*

Taylor series provide a way in which to approximate infinitely differentiable functions around a point. The Taylor series of an infinitely differentiable function $f(x)$ of a single variable around a point $x_0$ is

$$\begin{aligned} f(x) &= f(x_0) + \sum_{i \geqslant 1} \frac{1}{i!} (x - x_0)^i \left. \frac{d^i f(x)}{dx^i} \right|_{x = x_0} \\ &= f(x_0) + \sum_{i=1}^{k} \frac{1}{i!} (x - x_0)^i \left. \frac{d^i f(x)}{dx^i} \right|_{x = x_0} + r_{k+1}(x), \end{aligned} \tag{A.139}$$

where $r_{k+1}(x)$ is a remainder term.

Often, we assume that the remainder term can be ignored, focus on just the first three terms of (A.139), and write

$$f(x) \approx f(x_0) + (x - x_0) \left.\frac{df(x)}{dx}\right|_{x=x_0} + \frac{1}{2}(x - x_0)^2 \left.\frac{d^2 f(x)}{dx^2}\right|_{x=x_0}. \tag{A.140}$$

While a multivariate extension of (A.139) exists, it is a three term expansion, analogous to (A.140), that finds the most use in portfolio construction and risk management:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0) \nabla f(\mathbf{x})|_{\mathbf{x}=\mathbf{x}_0} + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)' \mathbf{H}(f(\mathbf{x}))|_{x=x_0} (\mathbf{x} - \mathbf{x}_0). \tag{A.141}$$

It is important to keep in mind that a Taylor series expansion of $f(\mathbf{x})$ is a *local* expansion, and is typically valid only in a small region centered at $\mathbf{x}_0$. As $\mathbf{x}_0$ changes, so does its Taylor series, as well as its gradient and its Hessian.

### *A.3.3 Convex and Concave Functions*

A function $f(x)$ of single variable is *concave* if for any $\alpha \in [0,1]$, and for all $x, y$,

$$f((1-\alpha)x + \alpha y) \geqslant (1-\alpha) f(x) + \alpha f(y). \tag{A.142}$$

.

The function $f(x)$ is *strictly concave* if the strict inequality $f((1-\alpha)x + \alpha y) > (1-\alpha) f(x) + \alpha f(y)$ holds for any $\alpha \in (0,1)$ and $x \neq y$.

A differentiable function $f(x)$ is (strictly) concave on an interval if and only if its derivative $f'(x)$ is (strictly) monotonically decreasing on that interval. If a function $f$ has a second derivative $f''(x)$ on an interval, then $f$ is (strictly) concave if and only if $f''(x)$ is (negative) non–positive on that interval.

If $f$ is concave and differentiable, then it is bounded above by its first–order Taylor approximation:

$$f(x) \leqslant f(x_0) + f'(x_0) \cdot (x - x_0) \; \forall x, x_0, \tag{A.143}$$

and has a global *maximum* that is achieved at one of the two boundaries of the interval or at a point $x_0$ for which $f'(x_0) = 0$.

A function $f(x)$ of single variable is *convex* if $-f(x)$ is concave, i.e. for any $\alpha \in [0,1]$, and for all $x, y$, $f((1-\alpha)x + \alpha y) \leqslant (1-\alpha) f(x) + \alpha f(y)$. The function $f(x)$ is *strictly convex* if $f((1-\alpha)x + \alpha y) < (1-\alpha) f(x) + \alpha f(y) \; \forall \alpha \in (0,1)$ and $x \neq y$.

A differentiable function $f(x)$ is (strictly) convex on an interval if and only if its derivative $f'(x)$ is (strictly) monotonically increasing on that interval. If a function $f$ has a second derivative $f''(x)$ on an interval then $f$ is (strictly) convex if and only if $f''(x)$ is (positive) non–negative on that interval.

If $f$ is convex and differentiable, it is bounded below by its first-order Taylor approximation:

$$f(x) \geqslant f(x_0) + f'(x_0) \cdot (x - x_0) \; \forall x, x_0, \tag{A.144}$$

and has a global *minimum* that is achieved at one of the two boundaries of the interval or at a point $x_0$ for which $f'(x_0) = 0$.

*Jensen's inequality*, which sees extensive use in continuous time finance, concerns the expected value of convex and concave functions of a random variable, states that for any random variable $Z$, and for any convex function $f$,

$$E\left[f\left(Z\right)\right] \geqslant f\left(E\left[Z\right]\right). \tag{A.145}$$

If $f$ is concave, the direction of the inequality is reversed.

Jensen's inequality is easily seen to be true by setting $x_0 = E\left[Z\right]$ in (A.143) and (A.144), and then taking the expectation of both sides.

### *A.3.4 Multivariate Convex and Concave Functions*

To extend the notions of convexity and concavity to functions of many variables, we need to define a *convex set*. A set $S$ is said to be convex if for any $\alpha \in [0,1]$, and for all $\mathbf{x}, \mathbf{y} \in S$, the line segment joining $\mathbf{x}$ and $\mathbf{y}$ lies within, or on the boundary of, $S$, i.e.

$$(1-\alpha)\mathbf{x} + \alpha\mathbf{y} \in S. \tag{A.146}$$

A multivariable function $f(\mathbf{x})$ that is defined on a convex set $S$ is said to be concave if it satisfies (A.142) with the scalars $x$ and $y$ being replaced by vectors $\mathbf{x}$ and $\mathbf{y}$. Replacing scalars with vectors allows us to similarly define strictly concave, convex and strictly convex functions by analogy to their definitions in Section (A.3.3).

A twice–differentiable function $f(\mathbf{x})$ is convex (strictly convex) if its Hessian, $\mathbf{H}(f(\mathbf{x}))$, is positive semi–definite (positive definite), and concave (strictly concave) if its Hessian is negative semi–definite (negative definite). Intuitively, a strictly convex function must lie above, and a strictly concave function must lie below, a hyperplane that is tangent to its surface at any point $\mathbf{x}_0$. Equation (A.141) makes clear that this can only be guaranteed for sufficiently small $\mathbf{x} - \mathbf{x}_0$ if the Hessian is positive definite (for convex functions) or negative definite (for concave functions).

### *A.3.5 Euler's Theorem for Homogeneous Functions*

A function $f(\mathbf{x})$ is said to be homogeneous of degree $k > 0$ if $f(c\mathbf{x}) = c^k f(\mathbf{x})\ \forall c > 0$.

Euler's theorem for homogeneous functions states that $f(\mathbf{x})$ is homogeneous of degree $k$ if and only if

$$f(\mathbf{x}) = \frac{1}{k}\sum_{i=1}^{n} x_i \frac{\partial f(\mathbf{x})}{\partial x_i}. \tag{A.147}$$

By way of example, $f(\mathbf{x}) = x_1^2 + x_2^2$ is homogenous of degree 2 because

$$\begin{aligned} f(c\mathbf{x}) &= c^2\left(x_1^2 + x_2^2\right) \\ &= c^2 f(\mathbf{x}) \end{aligned} \tag{A.148}$$

and it is easily seen that

$$\begin{aligned} f(\mathbf{x}) &= \frac{1}{2}(x_1 \cdot 2x_1 + x_2 \cdot 2x_2) \\ &= \frac{1}{2}\left(x_1 \frac{\partial f(\mathbf{x})}{\partial x_1} + x_2 \frac{\partial f(\mathbf{x})}{\partial x_2}\right). \end{aligned} \tag{A.149}$$

## A.4 Optimization

### A.4.1 The General Problem of Optimization

A very general statement of an optimization problem is

$$\begin{aligned} &\text{maximize} \quad f(\mathbf{x}) \\ &\text{subject to} \\ &\qquad g(\mathbf{x}) \geqslant 0 \\ &\qquad h(\mathbf{x}) = \mathbf{b} \\ &\qquad \mathbf{x} \in S. \end{aligned} \tag{A.150}$$

This problem is virtually limitless in both its scope and its difficulty: $f(\mathbf{x})$, $g(\mathbf{x})$ and $h(\mathbf{x})$ may be both nonlinear and discontinuous in their arguments, and $S$ may be bounded, non–convex, finite or countable (e.g. some or all the components of $\mathbf{x}$ may be restricted to be integers); $f(\mathbf{x})$ may have multiple local maxima that are not easily distinguished from its global maximum, and its evaluation may be numerically unstable and computationally demanding.

We therefore focus here only on the optimization of sufficiently smooth (i.e. twice differentiable) concave (or convex) functions defined on $\mathbb{R}^n$ with linear equality constraints. Fortunately, this covers many of the problems that actually arise in portfolio management.

### A.4.2 Optimizing Twice–Differentiable Convex Functions

Twice–differentiable convex (concave) functions with positive (negative) definite Hessians play a central role in portfolio construction as they are easily optimized when $S$, the convex set on which they are defined, is unbounded. Any extremum of such a function must necessarily be a minimum or a maximum, depending on whether the Hessian is positive or negative definite. The sole condition for an extremum is

$$\nabla f\left(\mathbf{x}_{opt}\right) = \mathbf{0}, \tag{A.151}$$

where $\nabla f(\mathbf{x})$ is defined in (A.129). If $\mathbf{H}(f(\mathbf{x}))$ is positive (negative) definite, $f$ is minimized (maximized) at $\mathbf{x}_{opt}$. Intuitively, this follows from (A.141): as the gradient is 0 at $\mathbf{x}_{opt}$, the linear term disappears, and only the quadratic term survives, so that the function must increase (decrease) as one moves away from $\mathbf{x}_{opt}$ in any direction according to whether $\mathbf{H}(f(\mathbf{x}))$ is positive (negative) definite.

### *A.4.3 The Method of Lagrange Multipliers*

We next extend our technique for optimizing twice–differentiable convex (concave) functions with positive (negative) definite Hessians to permit the inclusion of a single linear equality constraint of the form

$$\mathbf{a}'\mathbf{x} = b, \tag{A.152}$$

where $\mathbf{a}$ is a vector with constant real entries.

The method of *Lagrange multipliers,* which maps a constrained optimization to a related unconstrained optimization, is a powerful technique for solving problems such as this. It starts by constructing a new function, the *Lagrangian*, that combines the function and the linear constraint:

$$\mathcal{L}(\mathbf{x}, \theta) = f(\mathbf{x}) - \theta(\mathbf{a}'\mathbf{x} - b). \tag{A.153}$$

The parameter $\theta$ is known as a Lagrange multiplier, and at a stationary point of $\mathcal{L}$, we must have

$$\nabla\mathcal{L}(\mathbf{x}, \theta) = \mathbf{0}, \tag{A.154}$$

where

$$\nabla\mathcal{L}(\mathbf{x}, \theta) = \begin{bmatrix} \frac{\partial\mathcal{L}(\mathbf{x},\theta)}{\partial x_1} \\ \vdots \\ \frac{\partial\mathcal{L}(\mathbf{x},\theta)}{\partial x_n} \\ \frac{\partial\mathcal{L}(\mathbf{x},\theta)}{\partial\theta} \end{bmatrix}. \tag{A.155}$$

The equality constraint (A.152) must be satisfied at any stationary point of $\mathcal{L}(\mathbf{x}, \theta)$, as, if it is not, we can increase the value of $\mathcal{L}(\mathbf{x}, \theta)$ by choosing a different value of $\theta$, contradicting our assumption that it is a stationary point.

Observe that for any fixed value of $\theta$, the Hessian of $\mathcal{L}(\mathbf{x}, \theta)$ is identical to that of $f(\mathbf{x})$, as differentiating $\mathcal{L}(\mathbf{x}, \theta)$ twice with respect to $x_i$ and $x_j$ for $\theta$ fixed eliminates all the terms contributed by the linear constraint. In particular, if the Hessian of $f(\mathbf{x})$ is positive (negative) definite, so is that of $\mathcal{L}(\mathbf{x}, \theta)$. It follows that if the Hessian of $f(\mathbf{x})$ is negative definite, $f(\mathbf{x})$ achieves its constrained maximum at a stationary point of $\mathcal{L}(\mathbf{x}, \theta)$.

An identical argument allows for the constrained minimization of functions with positive definite Hessians, and this technique is easily extended to multiple linear equality constraints – we need only add one Lagrange multiplier per constraint, and ensure that the derivative of the Lagrangian with respect to each Lagrange multiplier is 0.

Once we allow for the inclusion of inequality constraints or non–linear (typically quadratic) constraints, the optimization problem becomes much harder, and we refer the reader to specialized textbooks on optimization such as Luenberger and Ye (2008) and Cornuéjols et al. (2018). Additionally, Kwan (2007), Kwan (2018) and Markowitz et al. (2020) provide useful expositions of optimization models that are of particular relevance to portfolio management, along with Python and R code, as well as Excel spreadsheets, to assist in their implementation.

## References

Cornuéjols, Gérard, Peña, Javier, and Tütüncü, Reha (2018). *Optimization Methods in Finance*. 2nd. Cambridge University Press. DOI: 10.1017/9781107297340.

Gentle, James E. (2017). *Matrix Algebra: Theory, Computations and Applications in Statistics*. 2nd ed. Springer.

Hager, William (1989). "Updating the Inverse of a Matrix". In: *SIAM Review* 31.2, pp. 221–239.

Hayashi, Fumio (2000). *Econometrics*. Princeton University Press.

Henderson, H. V. and Searle, S. R. (1981). "On Deriving the Inverse of a Sum of Matrices". In: *SIAM Review* 23.1, pp. 53–60.

Kwan, Clarence C. Y. (2007). "A Simple Spreadsheet-Based Exposition of the Markowitz Critical Line Method for Portfolio Selection". In: *Spreadsheets in Education* 3.2, pp. 253–280.

– (2018). "What really happens if the positive definiteness requirement on the covariance matrix of returns is relaxed in efficient portfolio selection?" In: *Financial Markets and Portfolio Management* 32, pp. 77–110.

Lu, Tzon-Tzer and Shiou, Sheng-Hua (2002). "Inverses of 2x2 Block Matrices". In: *Computers and Mathematics With Applications* 43.1-2, pp. 119–129.

Luenberger, David and Ye, Yinyu (2008). *Linear and NonLinear Programming*. 3rd ed. Springer.

Lv, Xiao-Guang and Huang, Ting-Zhu (2007). "A note on inversion of Toeplitz matrices". In: *Applied Mathematics Letters* 20.12, pp. 1189–1193.

Markowitz, Harry et al. (2020). "Avoiding the Downside: A Practical Review of the Critical Line Algorithm for Mean-Semivariance Portfolio Optimization". In: *Handbook of Applied Investment Research*. Ed. by John B. Guerard and William T. Ziemba. World Scientific Publishing Co. Pte. Ltd. Chap. 17, pp. 369–415.

Steele, Michael J. (2004). *The Cauchy–Schwarz Master Class*. Cambridge University Press.

Stevens, Guy (1997). *On the Inverse of the Covariance Matrix in Portfolio Analysis*. Tech. rep. Board of Governers of the Federal Reserve System.

– (1998). "On the Inverse of the Covariance Matrix in Portfolio Analysis". In: *Journal of Finance* 53.5, pp. 1821–1827.

Strang, Gilbert (2016). *Introduction to Linear Algebra*. 5th Ed. Wellesley – Cambridge Press.